Agent-based modeling of agricultural land-use and environmental policy

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Introduction

Land-use change is one of the main drivers of biodiversity loss, soil degradation, and changes in ecosystem functioning (Yin et al. [2020], Foley et al. [2005], Tilman et al. [2011]). Agricultural systems in particular must simultaneously provide food production, farmer income, and environmental public goods such as biodiversity, soil quality, and water regulation (Power [2011]. However, these objectives are often in conflict: intensive agriculture increases short-run production but may degrade environmental quality over time (Tanentzap et al. [2015], Tilman et al. [2002], Foley et al. [2005]). Designing policies that reconcile agricultural production and environmental sustainability is therefore a central problem in ecological economics and land system science (Lambin and Meyfroidt [2011]).

A key policy challenge is understanding how farmers respond to economic incentives aimed at promoting sustainable land use (Tanentzap et al. [2015]). Modern agriculture policies increasingly rely on environmental subsidies, conservation payments, and agri-environmental schemes to balance economic and environmental objectives (Pannell [2008]). However, the effectiveness of such policies depends on how farmers make decisions under heterogeneous conditions, including differences in land quality, production costs, and individual strategies (Pannell et al. [2006]). Understanding these behavioral responses is essential for predicting land-use outcomes and designing effective policy interventions.

Existing literature has approached this problem using both optimization models and agent-based models. Traditional optimization models often assume a representative farmer and, therefore, cannot capture heterogeneity or spatial interactions. In contrast, agent-based models (ABMs) have become an important tool for studying land-use systems because they allow heterogeneous agents, spatial interactions, and adaptive decision-making: land-use evolution becomes the result of decentralized decisions made by individual farmers operating under different constraints and incentives (Matthews et al. [2007], Dai et al. [2020]). These models have been used to study agricultural policy, biodiversity conservation and ecosystem services (Brady et al. [2012]). Recent research using agent-based models shows that agricultural policy can produce complex trade-offs between farm income, production, and environmental outcomes, and that policy design plays a crucial role in determining these outcomes (Coronese et al. [2023], Bayram et al. [2024], Baldi et al. [2025]).

Despite these advances, an important gap remains in understanding how different farmer decision rules interact with environmental subsidies to influence land-use allocation and end environmental dynamics within a heterogeneous landscape. In particular it remains unclear whether environmentally targeted subsidies can induce shifts away from intensive land use while maintaining farm-level profitability. This project addresses this gap by using an agent-based model in which heterogeneous farmers allocate land across different uses depending on land quality, production costs, and policy incentives. We compare three regimes: a productivist regime where farmers maximize output, a market regime where farmers maximize profit, and a policy regime where farmers maximize profit including environmental subsidies. By comparing these regimes, the model evaluates whether environmental subsidies can simultaneously support farm profitability and improve environmental outcomes. This approach follows the literature using agent-based land-use models to study agricultural policy and environmental outcomes (Matthews et al. [2007], Dai et al. [2020], Baldi et al. [2025]). This project contributes to this literature by constructing a simple theoretical agent-based model to study how environmental subsidies influence land-use decisions and environmental dynamics.

Methods

Modeling framework We developed a spatially explicit agent-based model (ABM) to study how heterogeneous farmers allocate land across alternative uses under different policy regimes. The model consists of a set of land plots and a population of farmers who make decentralized land-use decisions based on land-quality, production returns, and policy incentives. Land-use patterns emerge endogenously from these individual decisions.

Landscape The landscape consists of plots $k = 1, \ldots, K$, each characterized by an initial land quality $q_k \in [0, 1]$. Land quality is spatially auto-correlated, so that neighboring plots tend to exhibit similar levels of quality. Formally, let q_k be defined on a spatial grid, and assume

$$q_k = f(\varepsilon_k), \qquad \varepsilon_k \sim \mathcal{N}(0, 1),$$

where $f(\cdot)$ is a smoothing operator (e.g. a Gaussian kernel applied over neighboring plots), inducing correlation such that

$$\mathrm{Cov}(q_k, q_{k'}) = \sigma^2 \exp\left(-\frac{d(k, k')}{\ell}\right),$$

where $d(k, k')$ is the distance between plots k and k' and ℓ is a correlation length parameter. This formulation captures the empirical fact that land characteristics such as soil quality and environmental conditions are spatially clustered.

Farmers $i = 1, \ldots, N$ each control a subset K_i of plots, with $N \leq K$ and $\sum_i |K_i| = K$. Heterogeneity in farm size is introduced by allowing farmers to control different numbers of plots. Specifically, farm sizes $m_i = |K_i|$ are drawn from a lognormal distribution:

$$m_i \sim \mathrm{Lognormal}(\mu, \sigma^2),$$

which implies

$$\ln m_i \sim \mathcal{N}(\mu, \sigma^2).$$

This choice reflects the empirically observed right-skewed distribution of farm sizes, with many small farms and a few large ones. The m_i are then rescaled if necessary so that $\sum_i m_i = K$ and $m_i \geq 1$ for all i. In the baseline model, all plots are normalized to equal area, so larger farms are represented by ownership of more plots rather than larger individual plots.

To allocate farmers over this heterogeneous landscape, we assign ownership of plots to farmers in a way that is consistent with both farm-size heterogeneity and spatial structure. Let $m_i = |K_i|$ denote the number of plots controlled by farmer i. Conditional on the distribution of m_i, plots are assigned using a spatially clustered allocation rule: each farmer is first assigned an initial seed plot drawn uniformly at random, and additional plots are then allocated iteratively by selecting neighboring plots of already owned land until the farmer reaches m_i plots. Formally, if $\mathcal{N}(k)$ denotes the set of neighboring plots of k, then new plots are drawn from $\bigcup_{k \in K_i} \mathcal{N}(k)$ until $|K_i| = m_i$. This procedure generates spatially contiguous farm holdings and ensures that farmers operate on locally similar land, reflecting real-world patterns of land consolidation.

At each time step, farmers observe current land quality, environmental conditions, and policy incentives, and choose the land use of each plot. In the baseline model, these decisions are updated period by period; as a possible extension, adjustment costs or switching frictions may be introduced to limit unrealistic year-to-year reallocation.

Production Let $a_{i,k,t} \in \{I, O, S\}$ denote the land use chosen by farmer i on plot k at time t, where I denotes intensive agriculture, O organic agriculture, and S conservation/set-aside. Output on plot k at time t depends on current land quality and land use:

$$y(a_{i,k,t}, q_{k,t}),$$

with

$$y(I, q_{k,t}) > y(O, q_{k,t}) > y(S, q_{k,t}) = 0.$$

To capture how productivity depends on land quality, we specify yield functions in the following way:

$$y(a_{i,k,t}, q_{k,t}) = \alpha_{a_{i,k,t}} q_{k,t}^{\gamma_{a_{i,k,t}}},$$

Where $\alpha_{a_{i,k,t}}$ and $\gamma_{a_{i,k,t}}$ are positive parameters that determine the productivity and the responsiveness of the yields to the quality of the land.

Production Cost Intensive agriculture relies heavily on purchase inputs such as fertilizers and pesticides, while organic production requires greater labor and management efforts. Production cost also tend to rise with the decline of land quality, as there is an increase in input requirements on degraded soils (U.S. Department of Agriculture [2023], Durham and Mizik [2021]). So, baseline costs are defined as:

$$c(I, q_{k,t}) = c_I + \kappa_I(1 - q_{k,t})$$

$$c(O, q_{k,t}) = c_O + \kappa_O(1 - q_{k,t})$$

$$c(S, q_{k,t}) = c_S + \kappa_S(1 - q_{k,t})$$

Environmental dynamics Each plot has an environmental state variable $E_{k,t}$ representing soil quality / biodiversity index / ecological condition. Environmental quality evolves according to land use:

$$E_{k,t+1} = E_{k,t} + r_S \mathbf{1}\{a_{i,k,t} = S\} + r_O \mathbf{1}\{a_{i,k,t} = O\} - d_I \mathbf{1}\{a_{i,k,t} = I\}.$$

We assume

$$d_I > r_S > r_O \geq 0,$$

so that conservation generates the strongest environmental improvement, organic agriculture a smaller positive effect, and intensive agriculture degrades environmental quality. Land quality then evolves according to environmental quality:

$$q_{k,t+1} = q_{k,t} + \theta E_{k,t}.$$

$\theta > 0$ measures the sensitivity of land quality to environmental conditions and determines the speed at which ecological dynamics translate into changes in productive capacity. Since $q_{k,t} \in [0, 1]$, θ is chosen sufficiently small so that updates remain bounded in this interval (after truncation if necessary), and can be interpreted as a scaling parameter linking environmental quality units to land productivity.

Subsidy design. To study how policy design shapes land-use allocation, we consider three stylized subsidy designs: a differentiated practice-based subsidy,a conversion subsidy, and a results-based subsidy.

First, we consider a *practice-based subsidy* that pays different amounts depending on the practice implemented:

$$Sub_{i,t}^{P} = s_O \sum_{k \in K_i} \mathbf{1}\{a_{i,k,t} = O\} + s_S \sum_{k \in K_i} \mathbf{1}\{a_{i,k,t} = S\},$$

with $s_S > s_O > 0$ when conservation is intended to generate a larger environmental gain than organic production.

Second, we consider a *conversion-to-organic subsidy* that rewards the transition from intensive to organic agriculture. A simple specification is:

$$Sub_{i,t}^{C} = s_C \sum_{k \in K_i} \mathbf{1}\{a_{i,k,t-1} = I,\ a_{i,k,t} = O\},$$

where $s_C > 0$ is a one-period payment made when a plot is converted from intensive to organic use. A richer alternative, closer to multi-year conversion support, is:

$$Sub_{i,t}^{C,T} = s_C \sum_{k \in K_i} \mathbf{1}\{0 < t - t_{i,k}^{conv} \leq T_C\},$$

where $t_{i,k}^{conv}$ is the period in which plot k was most recently converted to organic production.

Third, we consider a *results-based subsidy* tied to realized ecological outcomes rather than declared practices. Let $E_{k,t}$ denote an observable ecological performance indicator on plot k at time t. A simple specification is:

$$Sub^{R}_{i,t} = \beta \sum_{k \in K_i} E_{k,t}, \qquad \beta > 0,$$

or, more conservatively,

$$Sub^{R,th}_{i,t} = \beta \sum_{k \in K_i} \mathbf{1}\{E_{k,t} \geq \bar{B}\}.$$

Farmer objectives Let $p > 0$ denote the output price and let $c_i(a_{i,k,t})$ denote the cost incurred by farmer i from assigning land use $a_{i,k,t}$ to plot k at time t. We consider three decision regimes:

1. Output maximization (productivist regime):

$$\max Y_{i,t} = \sum_{k \in K_i} y(a_{i,k,t}, q_{k,t}).$$

2. Profit maximization (market regime):

$$\max \pi^{M}_{i,t} = \sum_{k \in K_i} \left[p\, y(a_{i,k,t}, q_{k,t}) - c_i(a_{i,k,t}) \right].$$

3. Profit with environmental subsidy (policy regime):

$$\max \pi^{P}_{i,t} = \sum_{k \in K_i} \left[p\, y(a_{i,k,t}, q_{k,t}) - c_i(a_{i,k,t}) \right] + Sub_{i,t}.$$

Farmer heterogeneity. To preserve the agent-based character of the model while keeping the specification parsimonious, farmers differ primarily in plot endowments, farm size, and private net costs of assigning a given land use to a plot. We therefore use a single farmer-specific cost function $c_i(a_{i,k,t}, q_{k,t})$, interpreted broadly as the private net cost of assigning practice $a_{i,k,t}$ to plot k, including production-related costs and any private burden associated with adoption. To capture heterogeneity across farmers, we introduce technology-specific cost components $\eta_{i,I}$ and $\eta_{i,O}$ representing the differences in management ability, technology adoption, and opportunity costs under intensive and organic production. Farmer-specific costs are defined as:

$$c_i(I, q_{k,t}) = c(I, q_{k,t}) + \eta_{i,I}$$

$$c_i(O, q_{k,t}) = c(O, q_{k,t}) + \eta_{i,O}$$

$$c_i(S, q_{k,t}) = c(S, q_{k,t})$$

where

$$\eta_{i,I}, \eta_{i,O} \sim \mathcal{N}(0, \sigma^2_\eta)$$

We do not impose a fixed ranking between the costs of intensive and organic agriculture, since their relative magnitudes may vary across crops, technologies, and local conditions. We assume only that conservation/set-aside has a lower direct private cost than active production, so that:

$$c_i(S, q_{k,t}) < c_i(I, q_{k,t}), \qquad c_i(S, q_{k,t}) < c_i(O, q_{k,t})$$

As a possible extension, we may introduce farmer-specific responsiveness to policy incentives. Let $\rho_i \in [0, 1]$ denote the probability that farmer i updates land use in response to a change in relative net returns, so that lower values of ρ_i capture greater behavioral stickiness. In the baseline model, however, responsiveness is not imposed directly and instead emerges from heterogeneity in costs, land quality, and farm structure.

Simulation experiments. We will simulate the model under the three decision regimes. For each scenario we will compare:

- total agricultural output;
- mean farmer profit and the share of farmers above baseline profit;
- land-use shares (I, O, S);
- average environmental quality $\bar{E}_t$;
- spatial clustering of intensive versus conservation land.

Expected contribution. Our expectation is that, without policy, the model will generate over-allocation to intensive land use, especially on high-quality plots, leading to environmental decline over time. A sufficiently well-calibrated eco-subsidy should partially reverse this pattern by making conservation or lower-impact cultivation privately attractive. The main result of interest is not simply that "more subsidy improves the environment", but whether there is an intermediate region in which environmental indicators improve substantially while farm profits are maintained.

Possible extension. If time permits, we will extend the model in two directions. First, we will introduce social imitation, so that farmers may probabilistically copy the land-use choices of nearby successful farmers. Let $\mathcal{N}(i)$ denote the set of farmers located within a given spatial radius of farmer i, and let w_{ij} be proximity weights satisfying $w_{ij} \propto \exp(-d_{ij}/\lambda)$ and $\sum_{j\in\mathcal{N}(i)} w_{ij} = 1$, where d_{ij} is the distance between farmers i and j and λ governs the spatial decay of influence. Let $\bar{\pi}_{j,t}$ denote farmer j's recent average profit. Then farmer i updates by imitation with probability

$$P_{i,t}^{imit} = \rho_i \sum_{j\in\mathcal{N}(i)} w_{ij} \frac{\max\{\bar{\pi}_{j,t} - \bar{\pi}_{i,t}, 0\}}{1 + \max_{\ell\in\mathcal{N}(i)} |\bar{\pi}_{\ell,t} - \bar{\pi}_{i,t}|},$$

where $\rho_i \in [0, 1]$ captures farmer-specific behavioral responsiveness. Conditional on imitation, farmer i copies the land-use rule or plot-level allocation of a neighboring farmer j drawn with probability proportional to $w_{ij} \max\{\bar{\pi}_{j,t} - \bar{\pi}_{i,t}, 0\}$.

Second, we will allow for path dependence and threshold ecological dynamics. In particular, land quality may become difficult to restore once environmental conditions fall below a critical threshold. A simple specification is

$$q_{k,t+1} = q_{k,t} + \theta E_{k,t} - \delta\, \mathbf{1}\{E_{k,t} < \underline{E}\},$$

where $\underline{E}$ is a degradation threshold and $\delta > 0$ is an additional persistent loss in productive capacity incurred when the plot crosses that threshold. More generally, recovery may become weaker after collapse, for example:

$$q_{k,t+1} = q_{k,t} + \theta_1 E_{k,t}\mathbf{1}\{E_{k,t} \geq \underline{E}\} + \theta_2 E_{k,t}\mathbf{1}\{E_{k,t} < \underline{E}\}, \qquad 0 \leq \theta_2 < \theta_1,$$

so that ecological degradation generates hysteresis. These extensions would allow us to study whether temporary policy or imitation-induced shifts in behavior can prevent irreversible land degradation or, conversely, whether early intensive use can lock parts of the landscape into persistently low-quality states.

Results

Discussion

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Appendix

Table 1: Model Variables and Parameters

Symbol	Meaning
i	Farmer
k	Plot
$i = 1, ..., N$	Index of farmers
$k = 1, ..., K$	Index of land plots
t	Time step
K_i	Set of plots controlled by farmer i
$q_{k,t}[0, 1]$	Land quality of plot k at time t
$E_{k,t}$	Environmental quality (e.g., soil or biodiversity index) of plot k
$d(k, k')$	Distance between plots k and k'
m_i	$\|K_i\|$
l	Spatial correlation length parameter
$a_{i,k,t} I, O, S$	Land-use choice (Intensive, Organic, Set-aside)
$y(a_{i,k,t}, q_{k,t})$	Output (yield) as a function of land use and land quality
$\alpha_{a_{i,k,t}}$	Positive parameters that determine the productivity
$\gamma_{a_{i,k,t}}$	Elasticity of yield with respect to land quality
I	Intensive agriculture
O	Organic agriculture
S	Conservation/set-aside